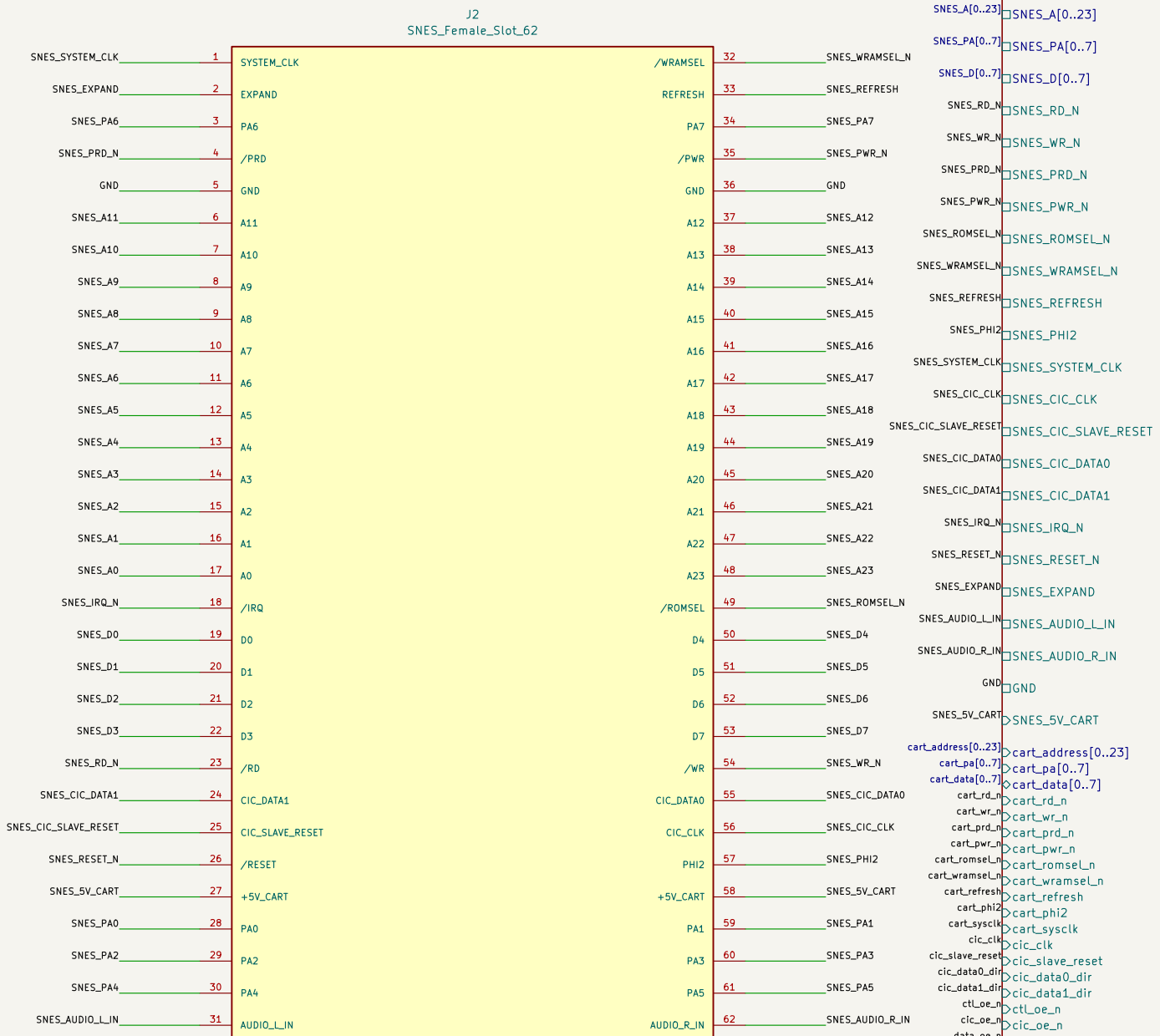
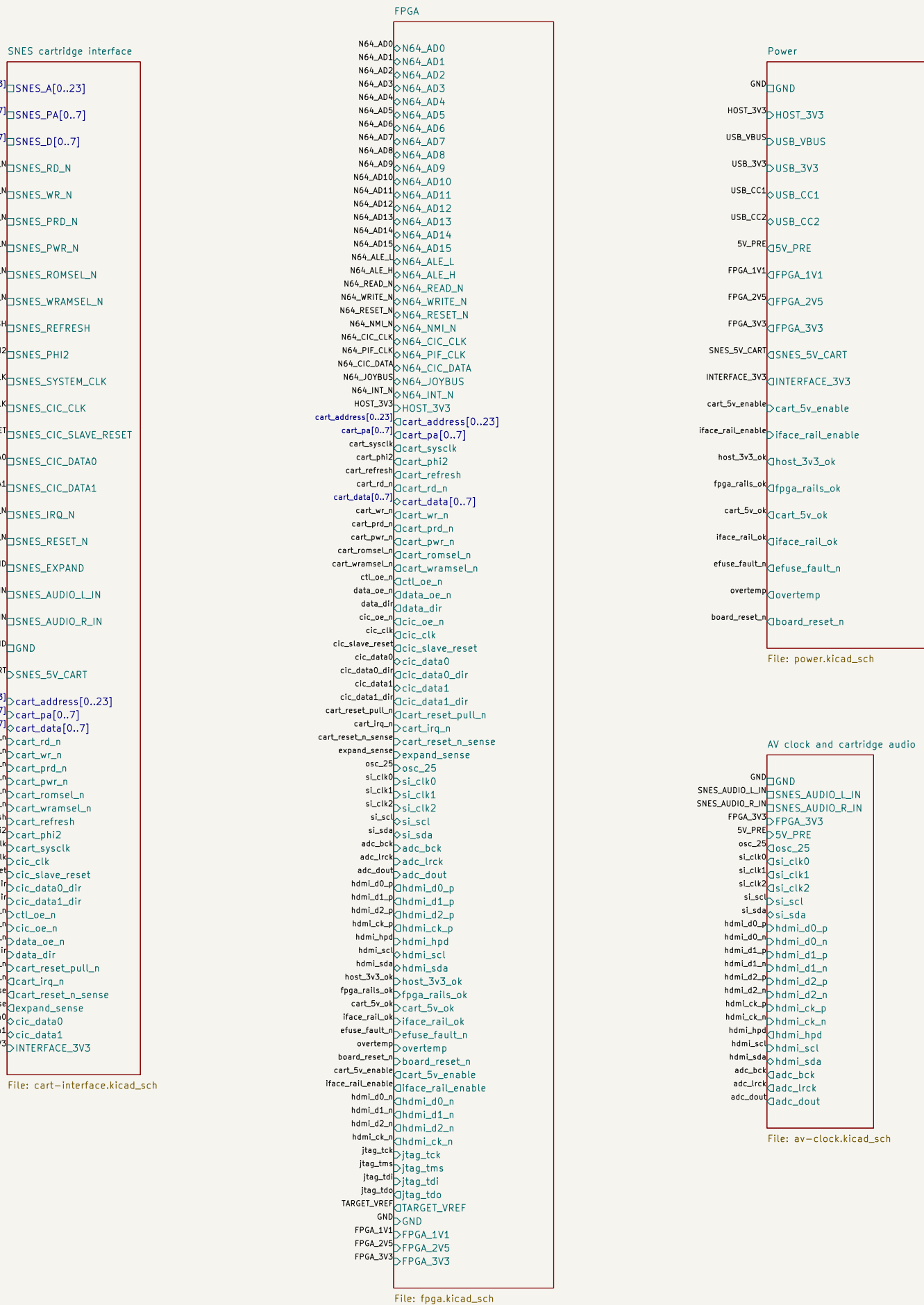


SNES / SUPER FAMICOM SOCKET

All 62 contacts retained, including analog cartridge audio



**SNES INTERFACE**  
5 V cartridge logic requires reviewed translation and isolation. SNES\_5V\_CART: protected switched supply, default OFF (pending). System RESET must support cartridge assertion and input sensing. CIC\_SLAVE\_RESET is separate; AUDIO\_L/R\_IN are analog inputs. No universal NC contacts. Full pin behavior is recorded in the CSV.



INTERFACE DRAFT ONLY – NOT FOR MANUFACTURE

Sheet: /

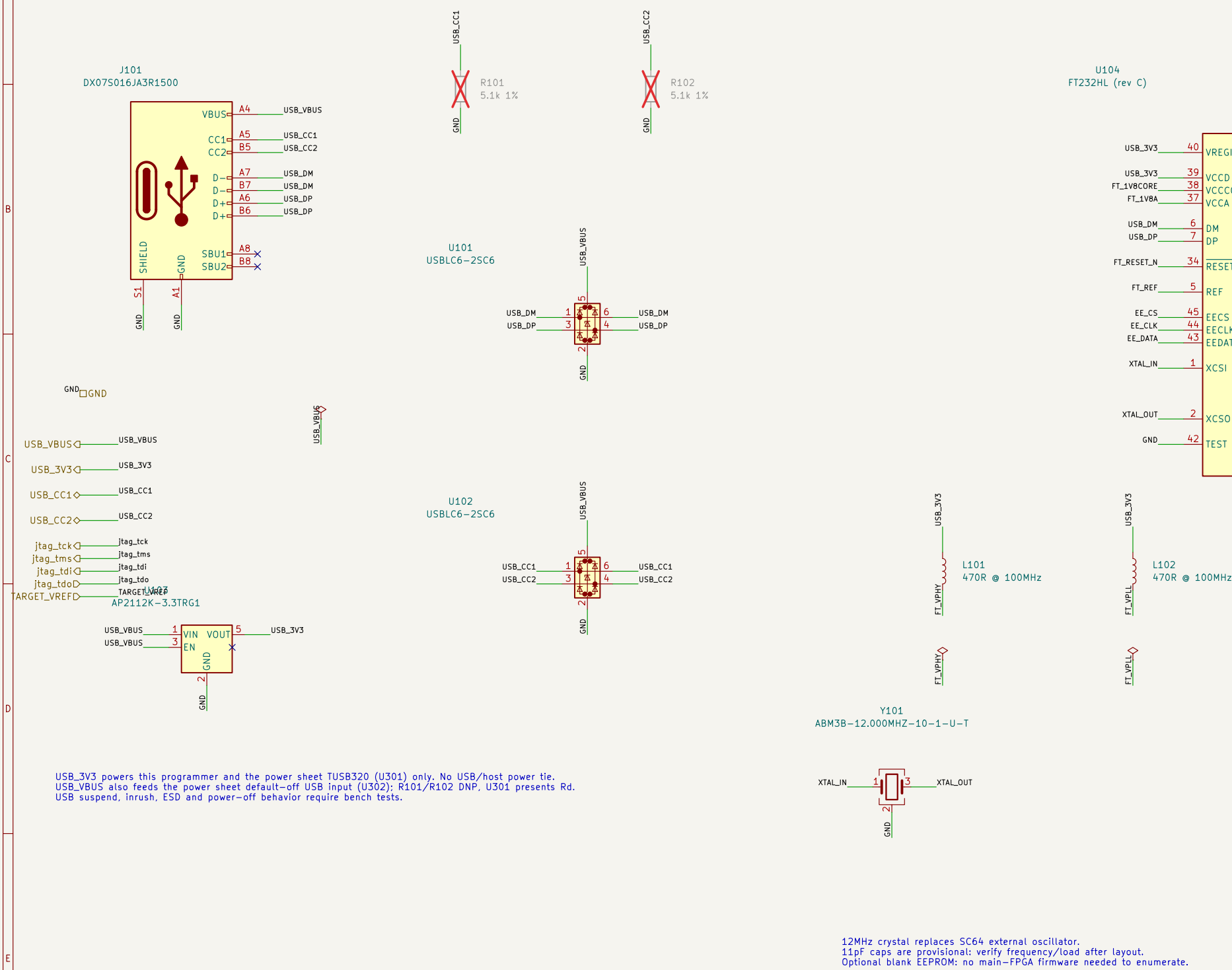
**Title: SN 64 – system schematic draft**

|                     |                  |             |
|---------------------|------------------|-------------|
| Size: A2            | Date: 2026-09-29 | Rev: 0.4-r3 |
| KiCad E.D.A. 10.0.6 |                  | Id: 1/6     |

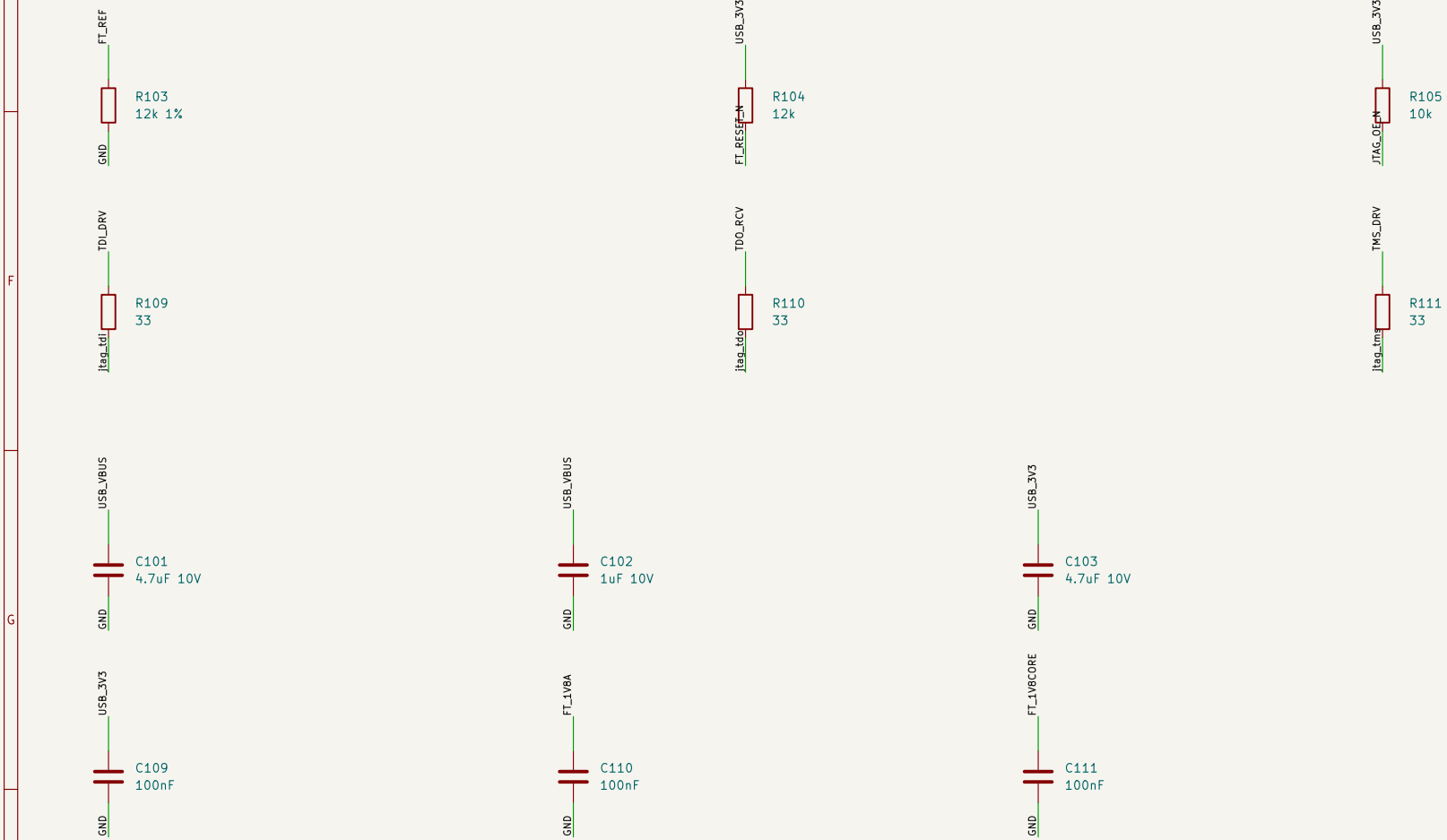
## SIDE USB-C: INITIAL PROGRAMMING + RECOVERY

USB 2.0 device; 5V input only. Side location is mandatory; exact PCB/enclosure placement remains open.

## FT232H HARDWARE BRIDGE

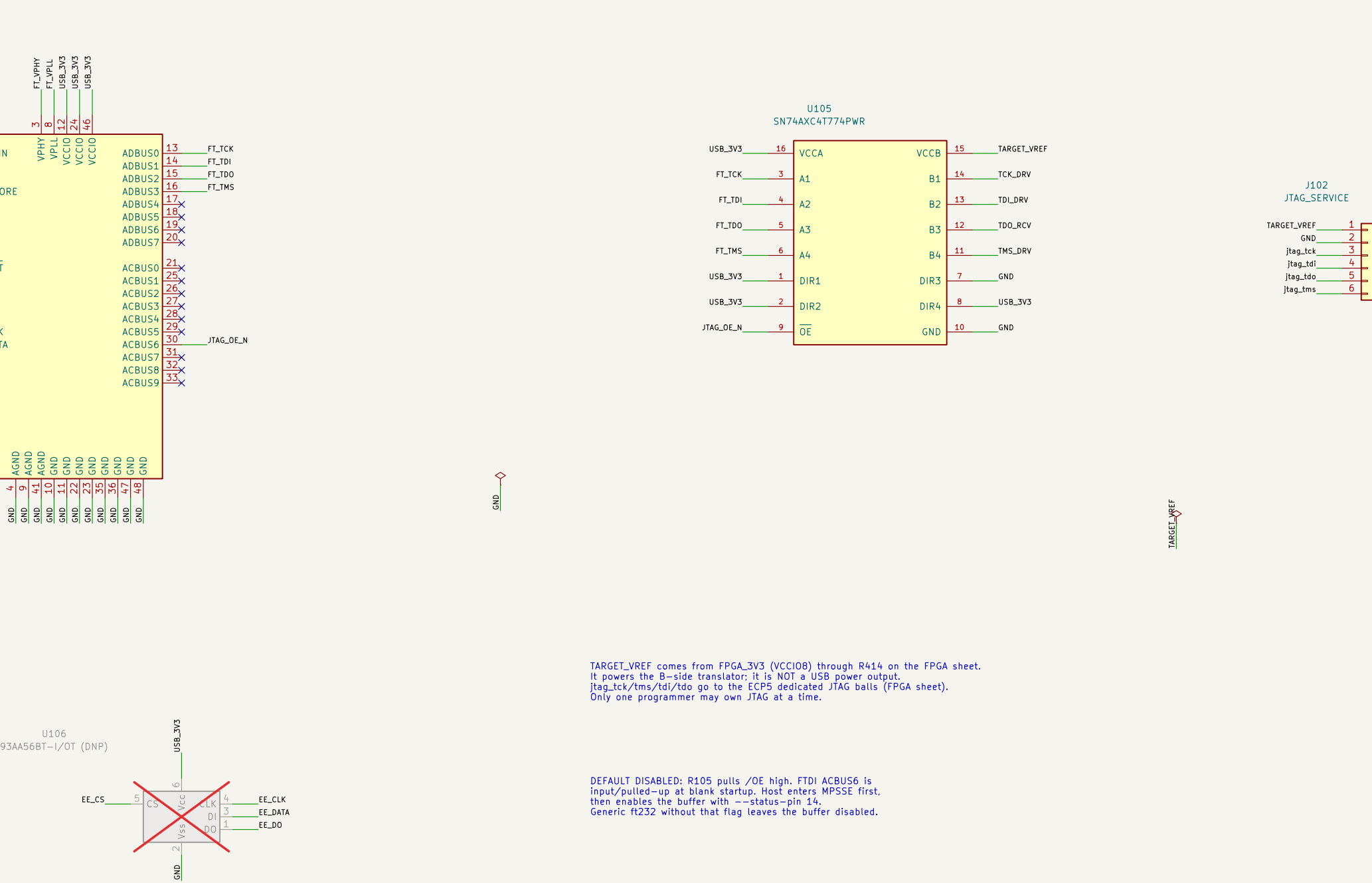


SUPPORT PASSIVES (place at their associated IC/pin; electrical grouping here is not PCB placement)



12MHz crystal replaces SC64 external oscillator.  
11pF caps are provisional; verify frequency/load after layout.  
Optional blank EEPROM: no main-FPGA firmware needed to enumerate.

## TARGET ISOLATION + SERVICE ACCESS



TARGET\_VREF comes from FPGA\_3V3 (VCCIOB) through R414 on the FPGA sheet. It powers the B-side translator; it is NOT a USB power output. jtag\_tck/tms/tdi/tdo go to the ECP5 dedicated JTAG balls (FPGA sheet). Only one programmer may own JTAG at a time.

DEFAULT DISABLED: R105 pulls /OE high. FTDI ACBUS6 is input/pulled-up at blank startup. Host enters MPSSE first, then enables the buffer with `--status-pin 14`. Generic ft232 without that flag leaves the buffer disabled.

DRAFT 0.2: USB programmer circuitry only. FPGA/storage choice, persistent-image map, system power, routing and physical validation are pending.  
Sources: SummerCart64 a1e7996 (CERN-OHL-S-2.0), FTDI FT232H v2.2, TI SN74AXC4T774, JAE drawing SJ122205. See docs/design/usb-programming-architecture.md.

DESIGN DRAFT – NOT FOR MANUFACTURE

SN 64

Sheet: /USB-C programmer/  
File: usb-programmer.kicad\_s

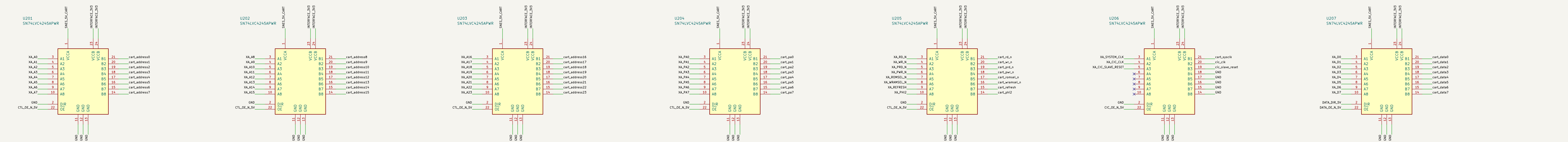
**Title: SN 64 – side USB-C programmer**

|                     |                  |              |
|---------------------|------------------|--------------|
| Size: A2            | Date: 2026-09-29 | Rev: 0.2-usb |
| KiCad E.D.A. 10.0.6 |                  | Id: 2/6      |

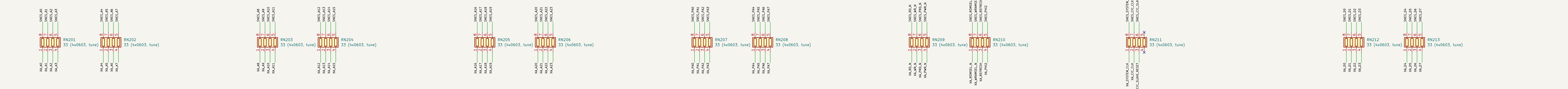
## SNES CARTRIDGE –SIDE TRANSLATION AND PROTECTION

SN74LVC4245A: A port = 5 V cartridge (VCCA = SNES\_V5\_CART), B port = 3.3 V FPGA (VCCB = INTERFACE\_V3V3); DIR/DE are referenced to VCCA (TI SCAS375K). DIR=L=H=A (FPGA drives socket); DIR=H=A=B; DE=H: Isolation.

U201–U206: console outputs, DIR strapped to GND (fixed FPGA→cartridge). U201–U205 share CTL\_DE\_N\_V3V (bridge ctl\_de\_n); U206 (SYSTEM\_CLK + CIC\_CLK + CIC\_SLAVE\_RESET only) has its own CIC\_DE\_N\_V3V (sn64\_top\_snes\_cic\_de\_n, enabled from the sequencer IFACE state).  
U207: DS–DT, DIR and DE both FPGA–controlled (bridge data\_dir / data\_de\_s). Unused B inputs tied to GND; unused A outputs no-connect.

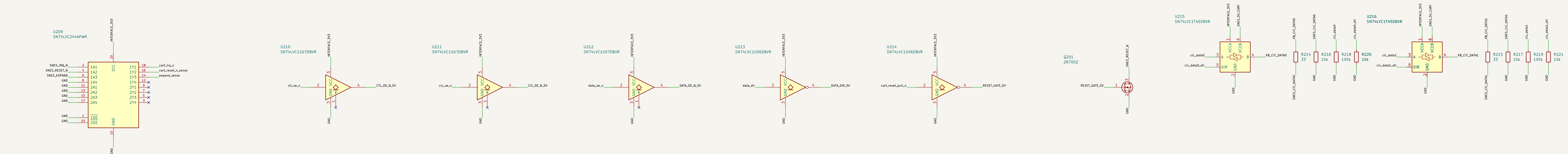


CARTRIDGE OUTPUT DAMPING: 33 R arrays at the translator A pins (source-series). Values are provisional; tune after SI measurement.



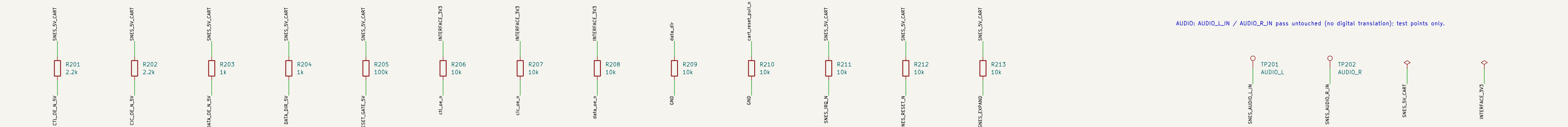
## SENSE + RELEASE PATHS

U209 (SN74LVC244A) at 3.3 V, 5.5 V-tolerant inputs with buff. recovers /IRQ, /RESET, EXPAND (CIC data pin via U215/U216).  
U210–U212 74LVC1G07 / U213 74LVC1G06, 3.3 V open-drain outputs (5.5 V tolerant, I<sub>OFF</sub> pulled to SNES\_V5\_CART).  
No 5 V boot-up reaches an FPGA pin, or FPGA unconfigured → every translator /OE released high (isolated).  
/RESET: Q201 open-drain sink, RESET\_GATE\_V3V pulled to 5 V; cartridge held in reset unless INTERFACE\_V3V3 is up AND the FPGA drives cart\_reset\_pull\_n=1.

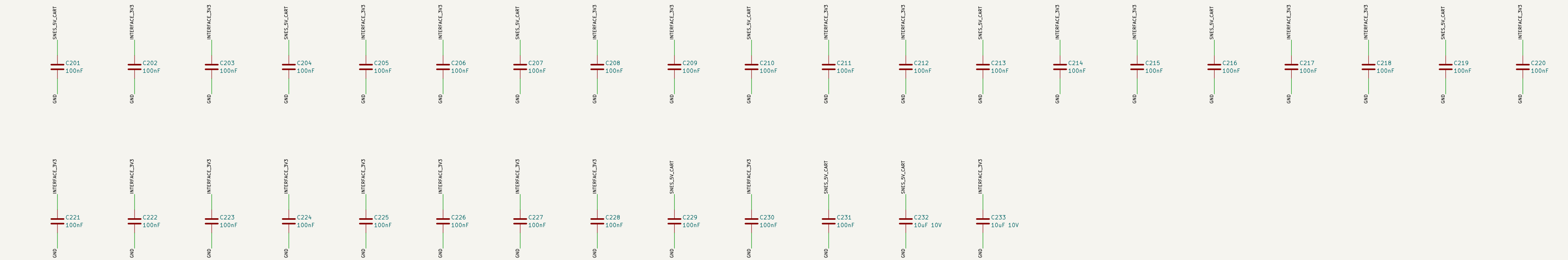


/OE and DIR edge timing: the bridge releases the data octet one A6.56 ns clock before turnaround. R203/R204 = 1k keep the passive rising edge short (about 10 pF load; tau < 10 ns). R201/R202 2.2k drive fno/rye /OE inputs where only permit-loss timing matters. Verify on hardware.

PULL/BIAS: /IRQ, /RESET, EXPAND 10k to cartridge 5 V (OpenSFC keeps pull-ups on IRQ/EXPAND); CIC\_DATA0/1 10k pull-downs (R216/R217). Values provisional.



DECOUPLING: one 100 nF per supply pin (place at the pin) plus one 10 uF bulk per rail.



DEBAT 0.51–cart: schematic circuit candidate only. No PCB, SI simulation, power sequencing test, ESD part or cartridge test has been performed.  
SNES\_V5\_CART and INTERFACE\_V3V3 are hierarchical inputs from the (future) power sheet: #U201/#U202 mark them externally supplied for ERC only.  
Open socket ESD array selection, final damping/pull values (CIC pull-downs value to confirm against a real key CIC).  
Source: TI SCAS375K (SN74LVC4245A), SCAS14AG (SN74LVC244A), SCE5296AG (SN74LVC1G07), SCE5296AG (SN74LVC1G06), SCE515N (SN74LVC1450).  
See docx/design/cart-interface-schematic.mxd.

## HIERARCHICAL PORTS – CARTRIDGE SIDE (root J2 net labels)

SNES\_A[0..23]→SNES\_A[0..23]  
SNES\_PA[0..7]→SNES\_PA[0..7]  
SNES\_DI[0..7]→SNES\_DI[0..7]  
SNES\_RD\_N→SNES\_RD\_N  
SNES\_WR\_N→SNES\_WR\_N  
SNES\_PRO\_N→SNES\_PRO\_N  
SNES\_PWR\_N→SNES\_PWR\_N  
SNES\_ROMSEL\_N→SNES\_ROMSEL\_N  
SNES\_WRAMSEL\_N→SNES\_WRAMSEL\_N  
SNES\_REFRESH→SNES\_REFRESH  
SNES\_PHI2→SNES\_PHI2  
SNES\_SYSTEM\_CLK→SNES\_SYSTEM\_CLK  
SNES\_CIC\_CLK→SNES\_CIC\_CLK  
SNES\_CIC\_SLAVE\_RESET→SNES\_CIC\_SLAVE\_RESET  
SNES\_CIC\_DATA0→SNES\_CIC\_DATA0  
SNES\_CIC\_DATA1→SNES\_CIC\_DATA1  
SNES\_IRQ\_N→SNES\_IRQ\_N  
SNES\_RESET\_N→SNES\_RESET\_N  
SNES\_EXPAND→SNES\_EXPAND  
SNES\_AUDIO\_L\_IN→SNES\_AUDIO\_L\_IN  
SNES\_AUDIO\_R\_IN→SNES\_AUDIO\_R\_IN  
GND→GND  
SNES\_V5\_CART→SNES\_V5\_CART

## PENDING PORTS – FPGA SIDE (bridge port names) = DEFAULT–OFF RAILS FROM FUTURE POWER SHEET

cart\_address[0..23]→cart\_address[0..23]  
cart\_pa[0..7]→cart\_pa[0..7]  
cart\_data[0..7]→cart\_data[0..7]  
cart\_rd\_n→cart\_rd\_n  
cart\_wr\_n→cart\_wr\_n  
cart\_pro\_n→cart\_pro\_n  
cart\_pwr\_n→cart\_pwr\_n  
cart\_romsel\_n→cart\_romsel\_n  
cart\_wramsel\_n→cart\_wramsel\_n  
cart\_refresh→cart\_refresh  
cart\_phi2→cart\_phi2  
cart\_sysclk→cart\_sysclk  
cic\_clk→cic\_clk  
cic\_slave\_reset→cic\_slave\_reset  
cic\_data0\_dir→cic\_data0\_dir  
cic\_data1\_dir→cic\_data1\_dir  
ctl\_de\_n→ctl\_de\_n  
cic\_or\_n→cic\_or\_n  
data\_or\_n→data\_or\_n  
data\_dir→data\_dir  
cart\_reset\_pull\_n→cart\_reset\_pull\_n  
cart\_irq\_n→cart\_irq\_n  
cart\_reset\_n\_sense→cart\_reset\_n\_sense  
expand\_sense→expand\_sense  
cic\_data0→cic\_data0  
cic\_data1→cic\_data1  
INTERFACE\_V3V3→INTERFACE\_V3V3

DESIGN DRAFT – NOT FOR MANUFACTURE

SN 64

Shiren /SNES cartridge interface/  
File: cart-interface.kicad\_sch

Title: SN 64 – SNES cartridge interface

Size: A1 Date: 2026–09–29

Kicad E.D.A. 10.0.6

Rev: 0.31–cart

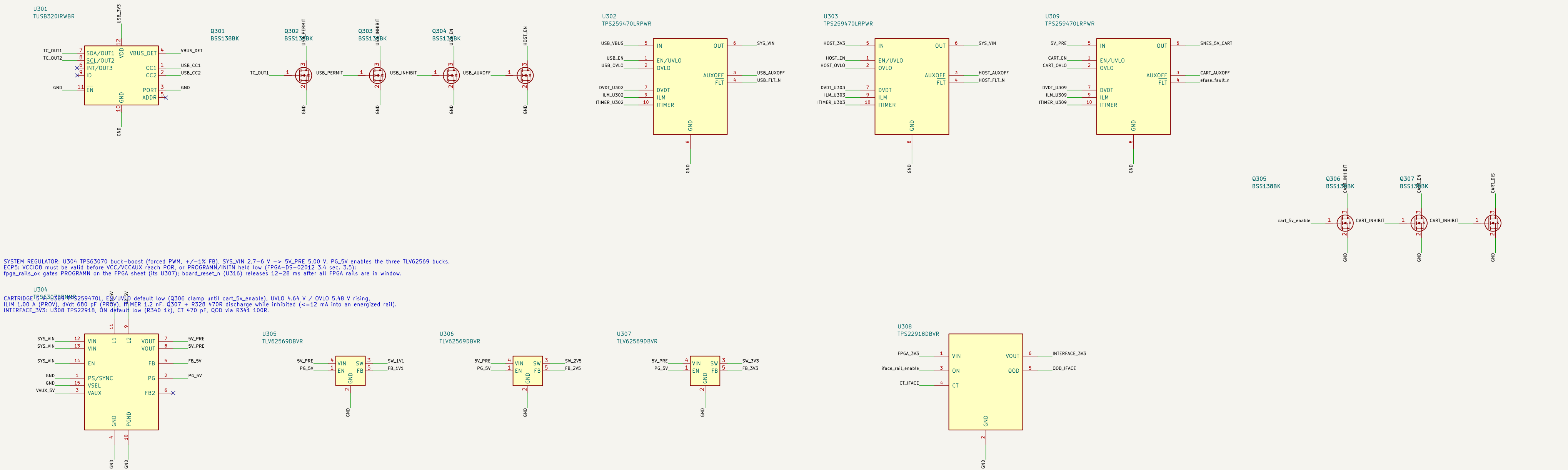
1/ 3/6



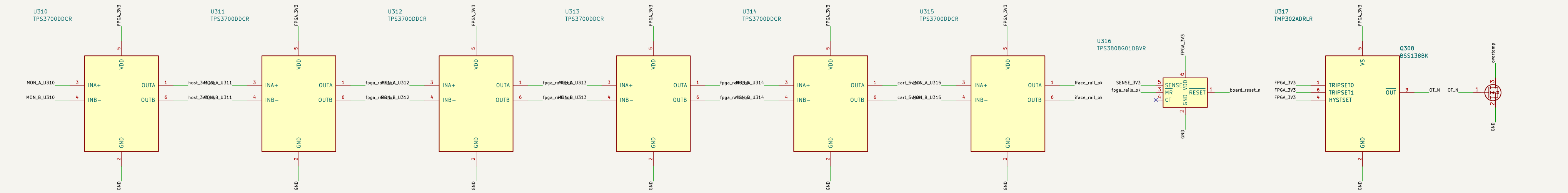
SN64 POWER – source selection, system regulator, FPGA rails, cartridge eFuse, interface rail, monitors

Implements docs/design/power-architecture.md math and sources in docs/design/power-schematic.md. HOST\_12V is not used on this sheet.  
Priority power max (1.5V3V3CVC figure 8-14): U302 (USB, priority) AUXOFF => Q304 puis U303 (HOST\_3V3, auxiliary) EN/VVLO low.  
Both eFuses have two reverse-current blocking, no source is ever directly paralleled onto SYS\_VIN.

USB permission (default off). U302 enables only when TUSB320 OUT1 is low (1.5 A or 3 A advertised) AND USB\_3V3 is present.  
TUSB320: PORT=GNP (OFF), ADDR open (GPIO mode). EN\_N=GNP, VDD=USB\_3V3 (power register PWR=LO6). The USB sheet 5.1k Rd must be DNP.



MONITORS (all powered from FPGA\_3V3, outputs pulled up to FPGA\_3V3 = 3.3 V-safe; supply loss reads NOT OK):  
U310 SYS\_VIN => host\_3v3\_ok | U311-U313 FPGA rails => fpga\_rail\_ok | U314 SNES\_SV\_CART (reset ide) => cart\_3v3\_ok |  
U315 INTERFACE\_3V3 => fpga\_rail\_en | U309 FLV via SV\_PFE divider => fpga\_rail\_en | U307 TP5301A + Q308 => overtemp (1 = not or sensor dead).



- GNP GND
- HOST\_3V3 HOST\_3V3
- USB\_VBUS USB\_VBUS
- USB\_3V3 USB\_3V3
- USB\_CC1 USB\_CC1
- USB\_CC2 USB\_CC2
- SV\_PFE SV\_PFE
- FPGA\_LV1 FPGA\_LV1
- FPGA\_2V5 FPGA\_2V5
- FPGA\_3V3 FPGA\_3V3
- SNES\_SV\_CART SNES\_SV\_CART
- INTERFACE\_3V3 INTERFACE\_3V3
- cart\_3v3\_enable cart\_3v3\_enable
- fpga\_rail\_en fpga\_rail\_en
- fpga\_rail\_ok fpga\_rail\_ok
- cart\_3v3\_ok cart\_3v3\_ok
- efuse\_fault\_n efuse\_fault\_n
- overtemp overtemp
- board\_reset\_n board\_reset\_n

DRAFT 0.1-power: schematic candidate only. No PCB, no measured load. Values marked PROV depend on unmeasured cartridge and FPGA current.  
Footprints for U302/U303/U309 (BPM), U304 (BPM) and U305-U308 are pending vendor initial-pattern review, they are left empty, not queried.

DESIGN DRAFT – NOT FOR MANUFACTURE

SN 64

Sheet: /Power/  
File: power.kicad\_sch

Title: SN 64 – Power

Size: A1

Date: 2026-09-29

Rev: 0.1-power

Kicad E.D.A. 10.0.6

16 / 5/6

SN 64 – CLOCKS, HDMI OUTPUT AND CARTRIDGE AUDIO ADC (DRAFT 0.1–av)

CLOCKS [docs/design/clock-plan.md], UPU SIS5351A-1C2 (Skynorks SIS5351-B-1 Rev. 1.3); CLK = NTC master clock at 74.6727272 MHz, CLK1 = PAL master, 21.28137 MHz, CLK2 = HDMI pin, 27.997947 MHz; programmed at power-on by the CPU's /snrclk/clock\_innvst over vscfclk+idv (CD address 0x00, MUX HCLK), VDD and VDDO both from FPGA\_SV33, CLK = 72: VDDO with or before VDD. One 100 nF per supply plan (sec. 7.4).

Crystal YX5321SL of oscillator ESFR <= 50 R, drive level up to 200 uW, CG <= 3 pF, SIS5351 part B needs ESR <= 150 R max drive level >= 100 uW, CL G 6.2 ohm.

VDDIO 25 MHz x3.3 CMOS oscillator (FPGA housekeeping, osc\_25); pin 1 enabled high tri-state, R706 or optional damping, Footprint pads differ slightly from the CJ005 series resistor (EM). Provisional.

CARTRIDGE AUDIO (SNES socket pins 11 AUDIO\_LIN / 62 AUDIO\_OUT), U703 PCN81B08 (T1 SLE5778) in i2S MASTER mode, 384 fs; MD1 = A (R708 to VDD), MDO = L, FMT = L (i2S 24-bit), X702 12.288 MHz = 384 x 32 kHz @ 10 = 32,000 kHz (own oscillator; asynchronous to the FPGA master rate-matrix). BCK = 64 fs = 2.048 MHz, outputs at VDDA\_3V3, SVCS (analog 5 V) = 5V1PR through FB701 + 10 uF + 100 nF; DOUT (digital) = FGPA3\_AVS/DGND/GND to GND at the device.

Input per channel: R711/R714 200 kΩ to GND = console load (OpenFSX CPU-CU-01 RS95/R96 200 R; its I<sub>UP</sub> up to 10 mA summing input in parallel).

Reference level: sDzSnEs RevE (C54344) at 3.3 V, 0.65 x VA typ.) ± 1.05 V<sub>RMS</sub> (L + 100 nF + 100 R) at pin 1 (largest digital cartridge supply) → 2.5 V<sub>P-P</sub>; ADC: 0.5000 to 0.5 V<sub>P-P</sub> at 0.6 x VCC = 3.0 V<sub>P-P</sub> (± 2.85 V<sub>RMS</sub> at 4.75 V).

Input impedance at the pin 200 [1]: 9.4k ± 10% (console); 200 R [1] (10k ± 196 R), 1 nF COT; 67.7 kHz anti-alias pole [Internal filter -3 dB at 1.3 MHz], 1 uF coupling; 2.6 Hz high-pass.

Reference levels: sDzSnEs RevE (C54344) at 3.3 V, 0.65 x VA typ.) ± 1.05 V<sub>RMS</sub> (L + 100 nF + 100 R) at pin 1 → -20.6 dBmHfz. Its SNES audio does NOT go to N64\_AUDIO/L/R (no requirement; see doc).



DRAFT 0.1--avoid schematic circuit candidate only. No PCB, SI simulation, HDMI compliance or audio measurement has been performed. FPGA\_3V3 and 5V\_PRE are hierarchical inputs from the power sheet (ERC origins there: #FLG304 and the TP563070 VOUT). Provisional: 0R series values, 200 R console--equivalent load (OpenSFC value, verify on SHVC--CPU--01). X7R 1 uF coupling, footprints of J701/X701/X702. Sources and SHA--256 of every dataset: libraries/av--provenance.json. See docs/design/av--clock--schematic.md.

DESIGN DRAFT – NOT FOR MANUFACTURE  
**SN 64**  
 Sheet: /AV clock and cartridge audio/  
 File: av-clock.kicad\_sch  
**Title: SN 64 – clocks, HDMI and cartridge audio**  
 Size: A2      Date: 2026-09-29  
 KiCad E.D.A. 10.0.6